

Cybernetic Security

Methods of network infrastructure protection

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Lecture Agenda

Securing of IP protocol

Network segmentation: VLAN

IDS, IPS

Network firewalls:

NGFW

web proxy

WAF

VPN

Securing IP protocol(s) – Risks

- Both IPv4 and IPv6 are not secure:
 - Risk of IP spoofing:
 - Sending packets with incorrect IP address of the sender:
 - Could pose a serious risks for networks, including:
 - Obfuscating endpoints into sending packets to different locations,
 - Convincing a victim host that a default gateway IP address is associated with incorrect MAC address (MITM attack).
 - Packet authentication is available but seldom used:
 - In IPSec, associated overhead is sometimes high (including configuration).
 - DHCP risk:
 - Counterfeit IP addresses could be assigned (by a rogue DHCP server).
 - MAC address assignment risks:
 - In IPv4, ARP is not secured at all,
 - In IPv6, Neighbor discovery can be authenticated but usually they are not.

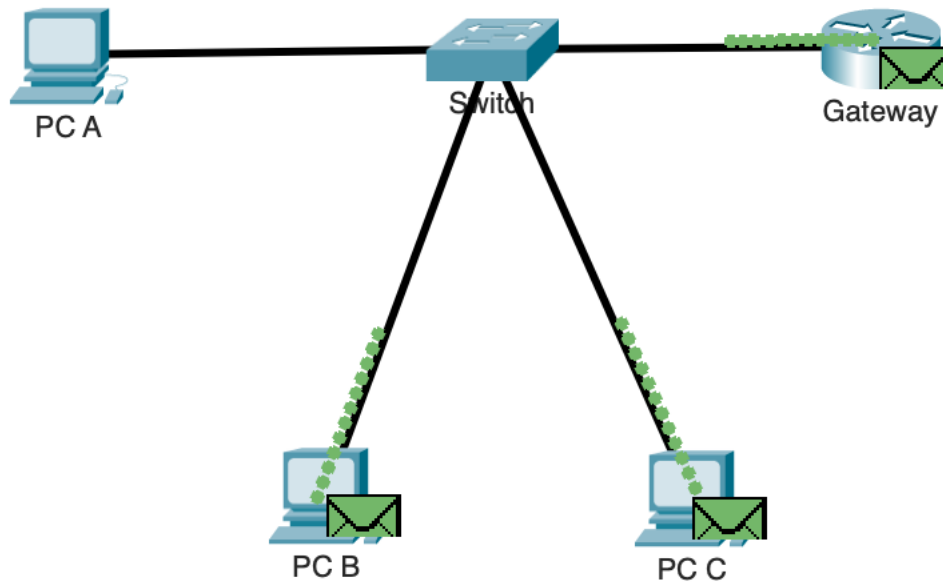
Securing IP protocol(s) – Prevention

- Various countermeasures are used to protect IP (L3) layer:
 - IP source guard on routers/L3 switches,
 - ARP inspection,
 - DHCP snooping,
 - ...

ARP Protection

- ARP is extremely vulnerable (no protection).
- ARP packets has no authentication:
 - Attacker just sends a fake ARP binding.
 - Pretending to be a response to previous request, or
 - Unsolicited (Gratuitous ARP).

ARP Vulnerability



- ARP requests are broadcasted (flooded).
- ARP protection is applied usually to L2 devices (switches).
- ARP protection techniques: various names (vendor-specific):
 - ArpON (ARP Handler),
 - ArpWatch,
 - Dynamic ARP Inspection (DAI).

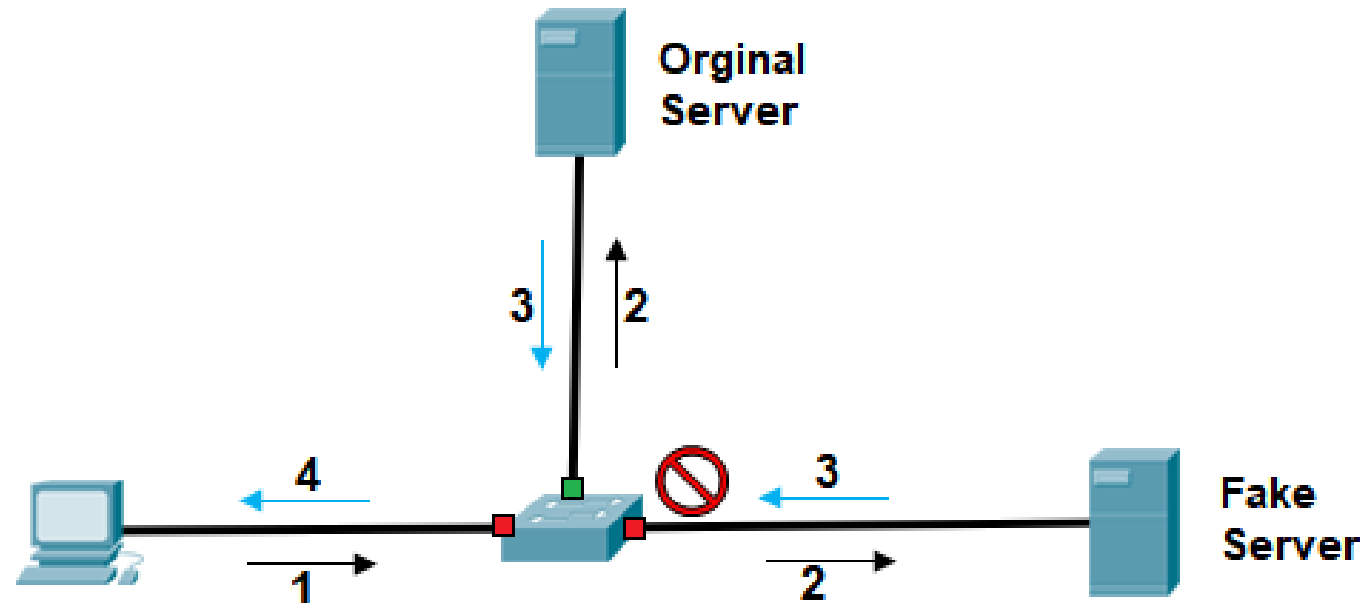
Dynamic ARP Inspection (DAI)

- ARP protection is based on storing ARP table on switches:
 - Where it is not stored normally.
- Together with DHCP snooping:
 - DHCP database is used as a storage for MAC address authentication.

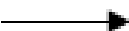
DHCP Securing – DHCP Snooping

- Multiple DHCP servers can operate in a network:
- No security is included in DHCP.
 - If a fake IP address is assigned to a client:
 - Client will communicate with other devices that the user intends.
 - MITM attack (communication eavesdropping) can happen.
- For an attacker, it is quite easy to install a fake DHCP server in a network:
 - e.g. as an unwanted service in a "zombie" PC,
 - difficult to detect and eliminate.
- DHCP snooping is a way how to suppress such fake DHCP servers:
 - through selecting ports from where DHCP offers can be passed (Trusted).

DHCP Snooping – Principle



Legend

	DHCP Client		L2 Switch		DHCPDISCOVER Path		Untrusted port
	DHCP Server		Link		DHCPOFFER Path		Trusted port

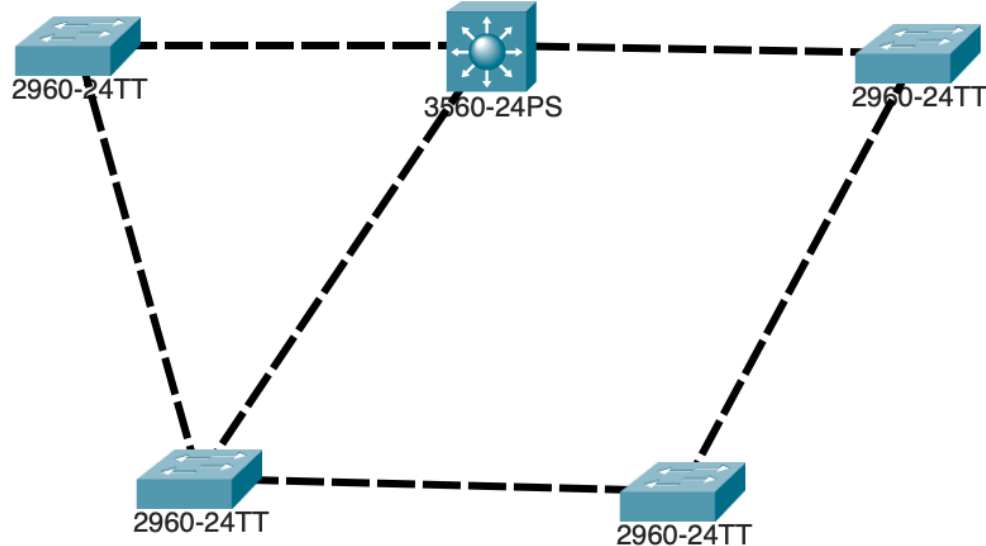
SLAAC Securing – RA Guard

- IPv6: DHCP is not the only way to assign IPv6 addresses dynamically.
- Stateless Automatic Address Configuration (SLAAC) can be used:
 - IPv6 prefix is distributed from routers via Router Advertisement messages.
 - RS/RA messages in ICMPv6.
- RA Guard:
 - Filtering RA messages on switches
 - Similar to DHCP snooping:
 - RA messages from non-router ports are discarded.

IP Source Guard

- Available on Cisco, Juniper and other devices.
- Operates in relation with DHCP snooping.
 - Every packet received from “untrusted” interface is checked:
 - IP address (source),
 - MAC address,
 - VLAN ID,
 - Interface,
 - Must match an item in DHCP database.
 - Otherwise, the packet is discarded.
 - Untrusted interface: most access interfaces.
 - Look one slide back for illustration.

Segmentation of Networks



“Flat” network:

- Only L2 interconnections,
- When growing, it becomes difficult to manage.
 - Complex STP.
- L2 segmentation into Virtual LANs:
 - VLANs are logically separated.
 - Load-balancing across multiple physical connections:
 - Various STP trees for different VLANs.
 - Inter-VLAN communication is possible via L3 devices:
 - Here, e.g. L3 switch (via routing features).

VLAN Benefits

- VLANs are used in **each** corporate network.
- Logical separation of traffic:
 - Ordinary user traffic:
 - with optional further separation e.g. based on organizational units.
 - Guest VLAN,
 - Management VLAN:
 - Not accessible for ordinary user traffic,
 - ...
- Separated traffic can be controlled easily using a router (L3 switch): ACL etc.

VLAN Potential Pitfalls and Issues

- Additional configuration necessity:
 - Without VLANs, switch configuration is almost unnecessary.
- Automatic port configuration:
 - DTP (dynamic trunk protocol) – potential security risk
- Double tagging is sometimes necessary:
 - Additional complexity for management.

Network Firewall

■ Basic approaches:

- Packet filtering:
 - Rules applied to every packet the same way:
 - Regardless of any previous communication:
 - Cannot handle e.g. packets belonging to previously established TCP connection differently.
 - Pro: low resource consumption (easy filtering, even can be hardware-accelerated).
 - Adversaries:
 - only very simple attacks can be avoided this way.
 - More complex policies cannot be implemented (e.g. allowing of packets responding to previous allowed traffic).
- Stateful packet filtering:
 - Added stateful distinguishment (e.g. filtering can differ according to previous logging in/connection establishment.
- NAT is also included (optional).

NextGen Firewall (NGFW)

- Additional features to packet filtering in traditional firewalls:
 - Complex L7 inspection:
 - www inspection,
 - Primary focus.
 - HTTPS decryption is required.
 - SSH, FTP, SMB inspection,
 - Antivirus/antimalware,
- Includes traditional firewall's packet filtering but
 - Builds a lot of features above it.
 - Uses strong identity management services:
 - The protection mechanisms are user-aware.

Intrusion Detection & Prevention

■ Intrusion Detection System (IDS):

- Network-Based appliance (service) detecting potential risky activities:
 - Complex detection of advanced attacks (APT).
 - Dynamic (self-adaptive) set of rules to detect intrusions (attacks)
 - Does not block the detected traffic (IDS is not in front of firewall but aside).

■ Intrusion Prevention System (IPS):

- Network-Based service detecting & blocking potential risky activities:
- Evolution of IDS.
- Located just behind a central firewall.

Web Traffic Control

■ Web Proxy:

- Web client protective component (centralized):
 - Collects HTTP(S) requests from web clients in the protected network, and
 - Resends them to target web servers on behalf of individual clients.
- Re-encryption of HTTPS traffic must be performed:
 - Including certificate management and password (identity) management.

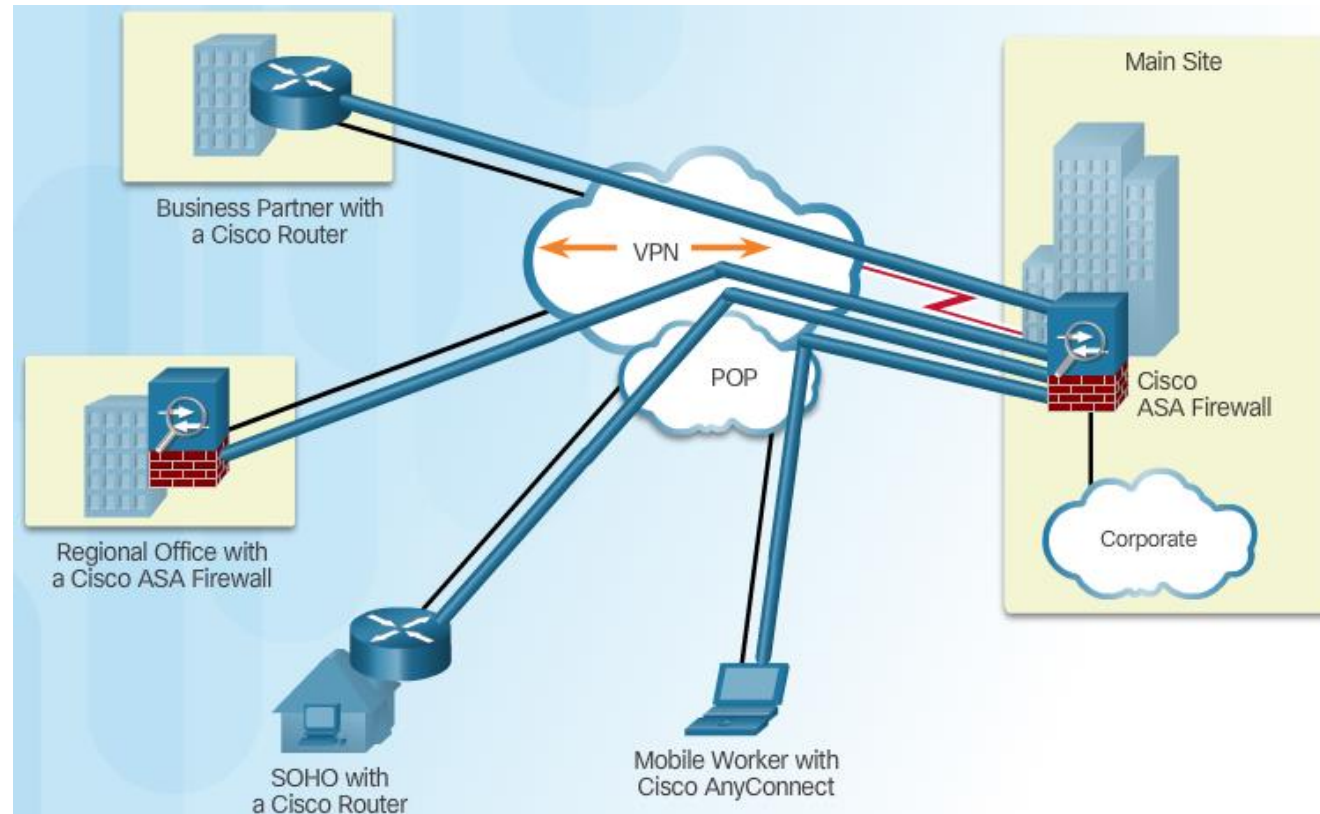
■ Web Application Firewall (WAF):

- Specialized device/SW component used in web/app servers:
 - To increase the security of web services provided.

Virtual Private Networks (VPN)

VPN Benefits:

- Cost Savings
- Security
- Scalability
- Compatibility
- ...



VPN Protocols

IPSec VPN

- L3 encryption via IPSec
- Often used for site-to-site VPN

SSL-based VPN

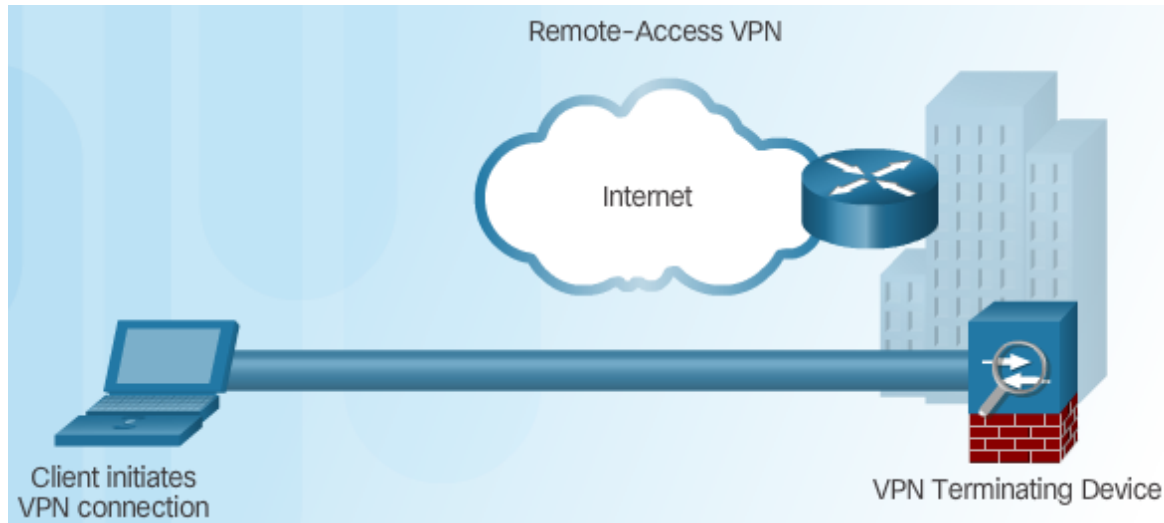
- L4/L7 encryption via TLS
- SSL is TLS predecessor
- Widely used in www browsers
- Easier to use for remote access VPN than IPSec VPN
- Sometimes less secure

L2TP VPN

- Combined with IPSec for data encryption (RFC 3193)

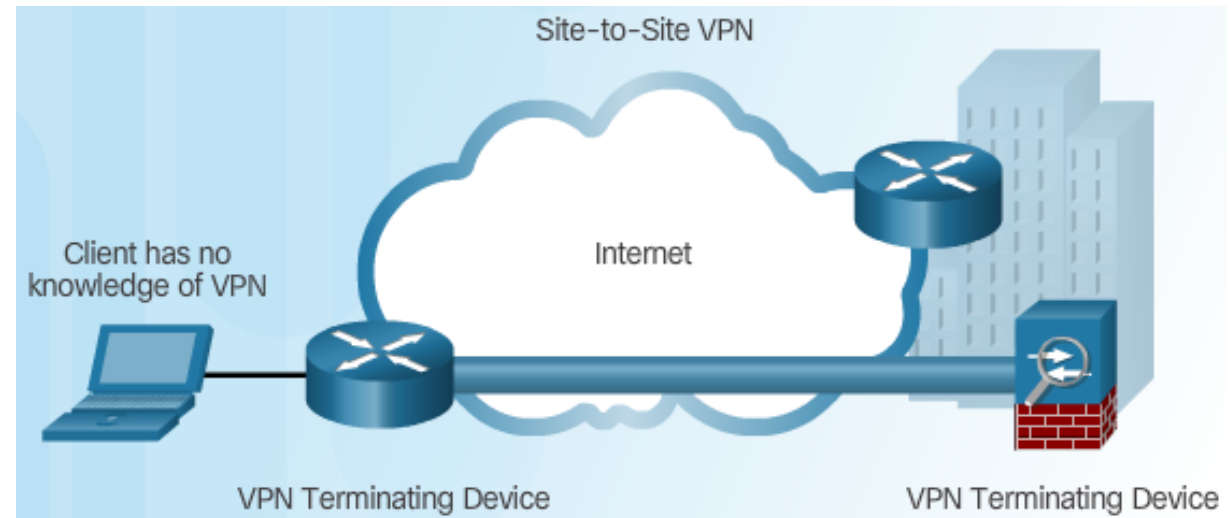
PPTP VPN - Obsolete

VPN Topologies

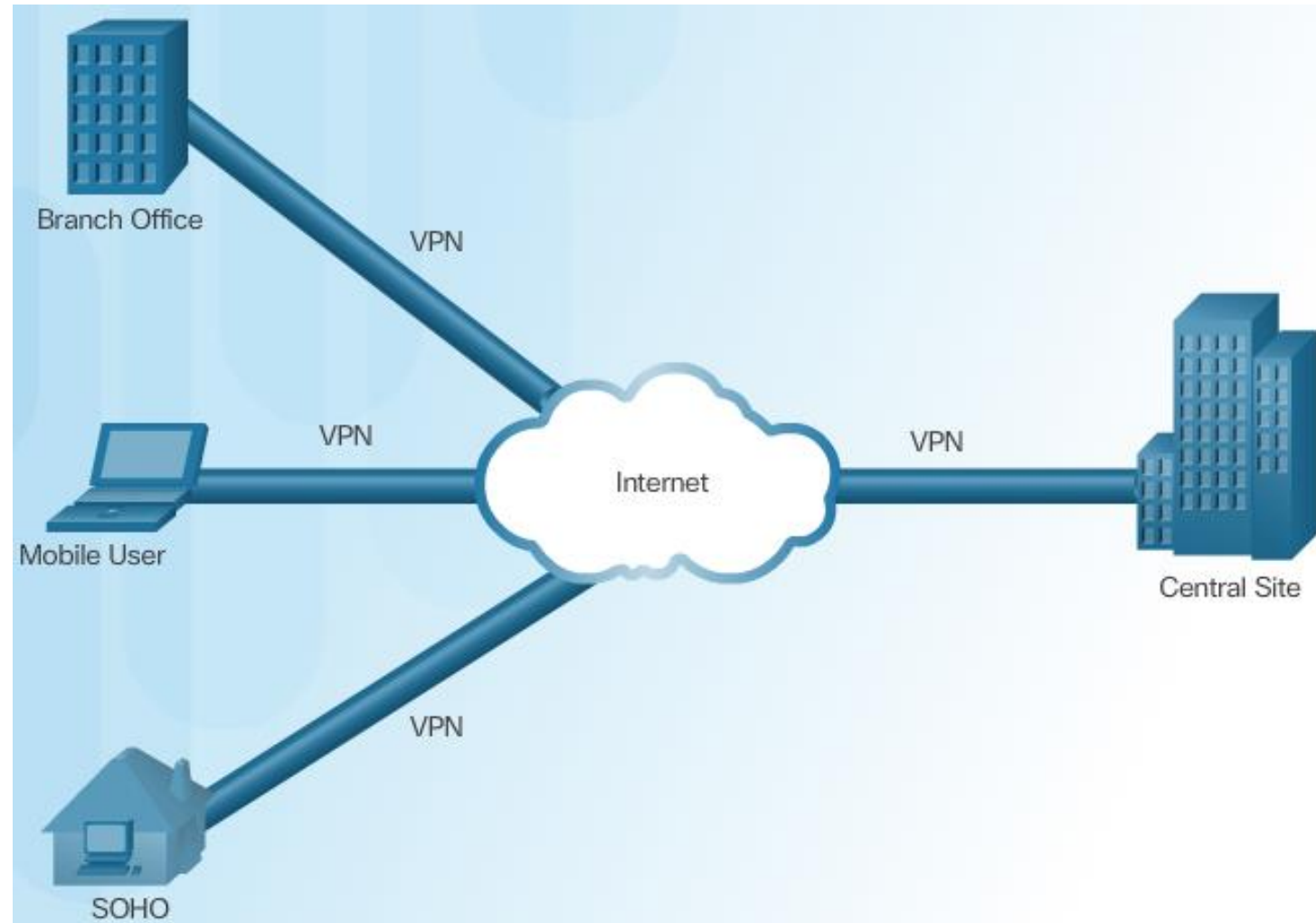


Remote-Access VPN

Site-to-Site VPN
Access



IPSec VPN

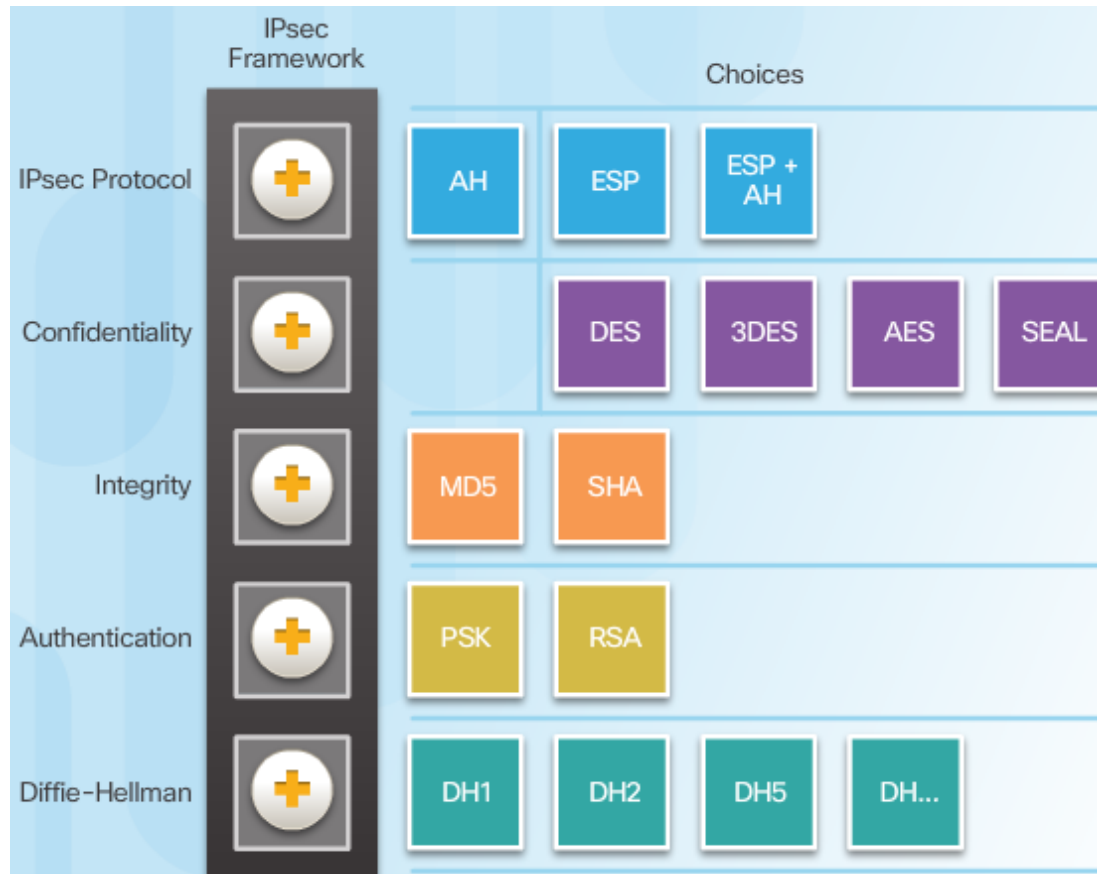


IPSec VPN Components

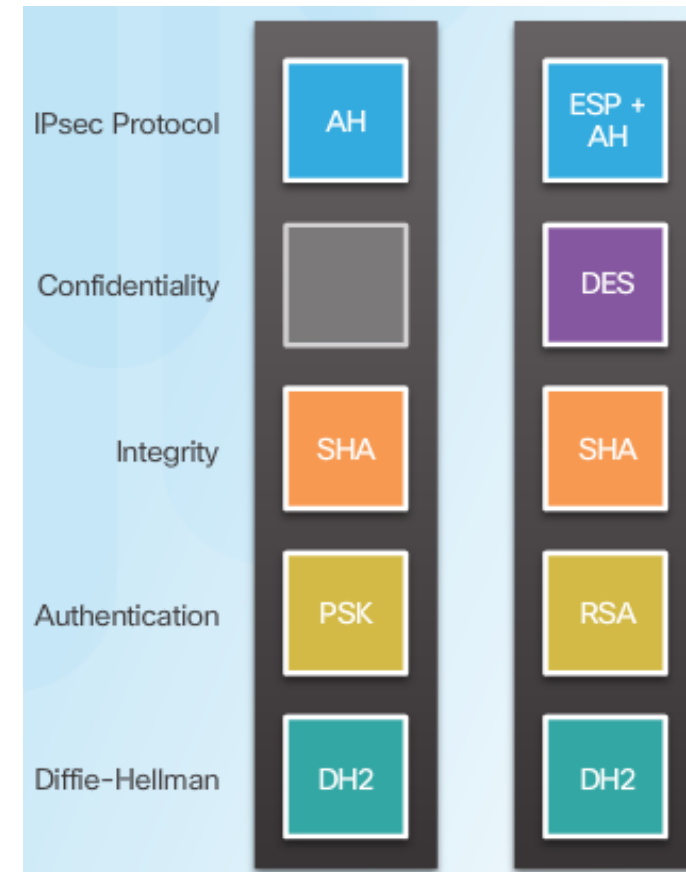
- IPsec protocol and its basic functions.
- AH and ESP protocols.
- IKE protocol.

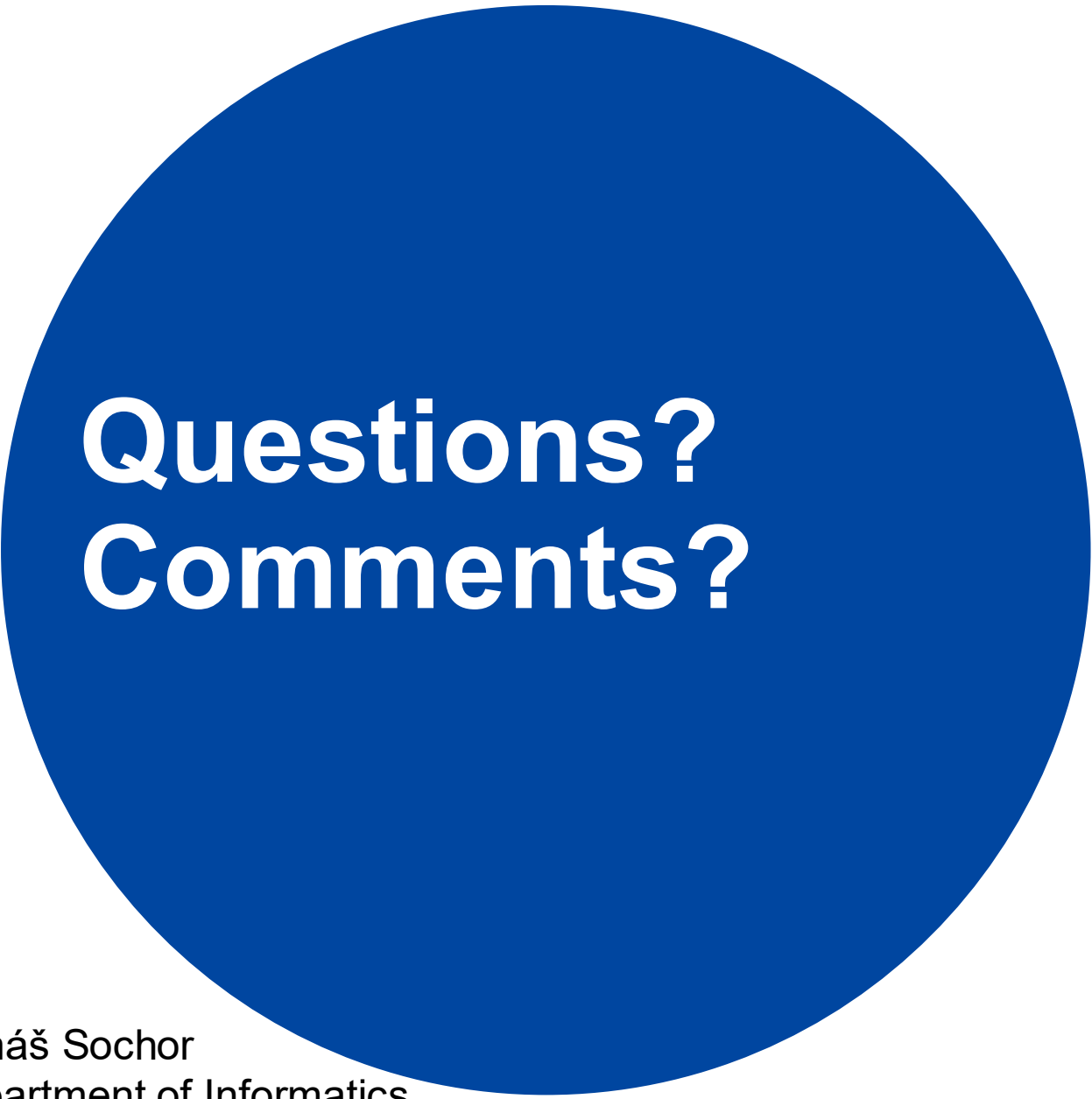
IPSec VPN Components

IPsec Framework



IPsec Implementation Examples





Questions? Comments?

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Thank you for your attention

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